

Q1

of poles: 6 $\Rightarrow$ pole pair $p = 3$

of slots: 72 (Q) 3 phase, double layer

Full Pitched:

$$\text{slot angle } \alpha = \frac{2\pi}{(Q/p)} = 15^\circ$$

$$q = \frac{Q}{2 \cdot m \cdot p}, \quad m = \# \text{ of phase} \Rightarrow q = 4$$

$$\text{Distribution factor } k_d = \frac{\sin(q \cdot \frac{\alpha}{2})}{q \cdot \sin(\frac{\alpha}{2})} = 0,957$$

$$\text{Pitch factor } k_p = \sin(\frac{\lambda}{2}), \quad \lambda: \text{coil pitch}$$

$$k_p = \sin(\frac{\pi}{2}) = 1$$

$$\text{Winding factor } k_w = k_p \cdot k_d = 0,957$$

for 3rd harmonics:

$$k_d = \frac{\sin(nq \frac{\alpha}{2})}{q \cdot \sin(n \frac{\alpha}{2})} = 0,653 \quad k_p = \sin(\frac{3\pi}{2}) = -1 \Rightarrow k_w = -0,653$$

for 5th harmonics:

$$k_d = 0,205 \quad k_p = 1 \Rightarrow k_w = 0,205$$

Winding Diagram: Slots per pole pairs: $72/3 = 24$

Q

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A_1^+	A_2^+	A_3^+	A_4^+	C_1^-	C_2^-	C_3^-	C_4^-	B_1^+	B_2^+	B_3^+	B_4^+	A_1^-	A_2^-	A_3^-	A_4^-	C_1^+	C_2^+	C_3^+	C_4^+	B_1^-	B_2^-	B_3^-	B_4^-

11/12 short - pitched :

k_d is same above, $k_d = \frac{\sin(n\frac{\alpha}{2})}{n \cdot \sin(\frac{\alpha}{2})}$

$k_p: \lambda = 165^\circ \Rightarrow k_p = \sin(\frac{\lambda}{2}) = 0.991$

$k_w = k_p \cdot k_d = 0.991 \cdot 0.957 = 0.948$

for 3rd harmonics:

$k_d = 0.653, k_p = 0.991 \Rightarrow k_w = 0.647$

for 5th harmonics:

$k_d = 0.205, k_p = 0.991 \Rightarrow k_w = 0.203$

Winding Diagram :

Q

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A_1^+	A_2^+	A_3^+	A_4^+	C_1^-	C_2^-	C_3^-	C_4^-	B_1^+	B_2^+	B_3^+	B_4^+	A_5^-	A_6^-	A_7^-	A_8^-	C_5^+	C_6^+	C_7^+	C_8^+	B_5^-	B_6^-	B_7^-	B_8^-
A_6^+	A_7^+	A_8^+	C_5^-	C_6^-	C_7^-	C_8^-	B_5^+	B_6^+	B_7^+	B_8^+	A_1^-	A_2^-	A_3^-	A_4^-	C_1^+	C_2^+	C_3^+	C_4^+	B_1^-	B_2^-	B_3^-	B_4^-	A_5^+

Q2

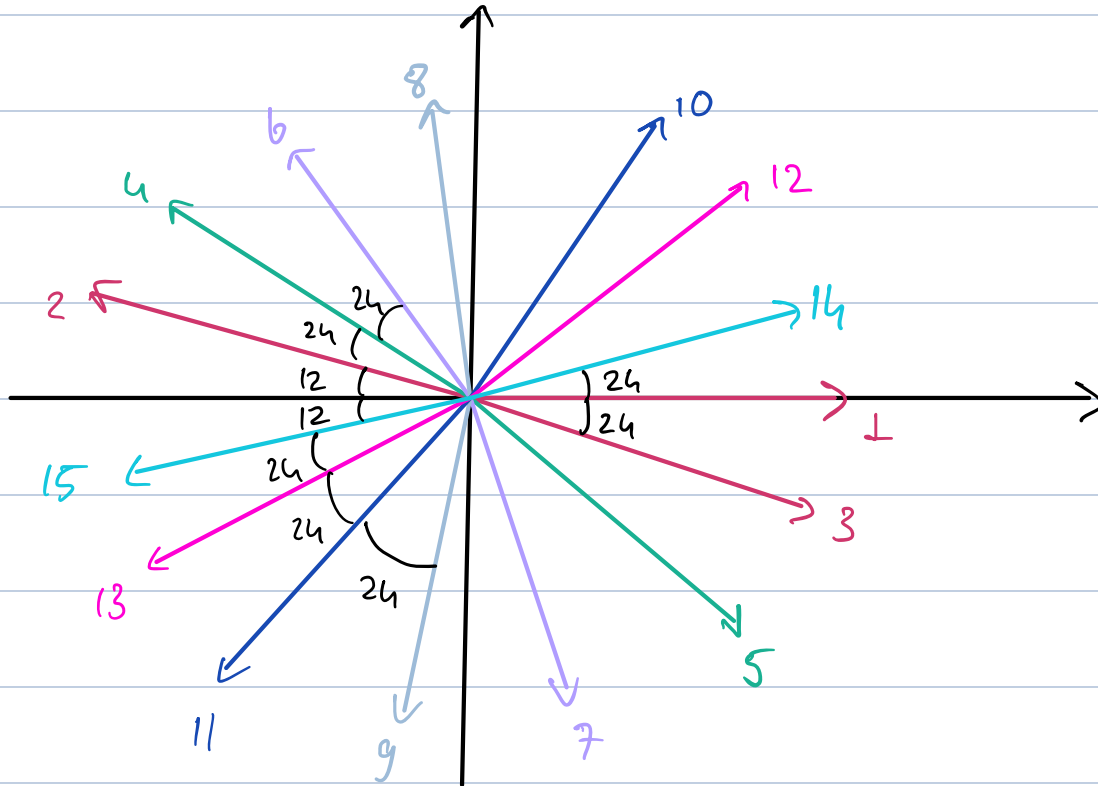
Number of slots : 15 (Q)

of poles : 14 $\Rightarrow p = 7, m = 3$

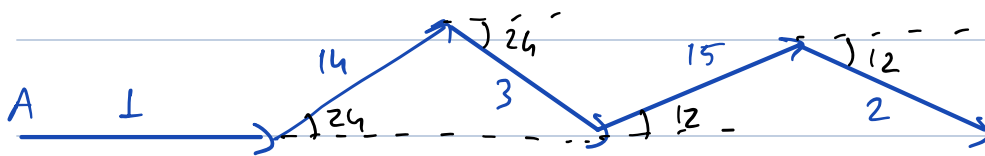
$$q = \frac{Q}{p \cdot m} = \frac{5}{7}$$

$$\alpha_0 = 360 \cdot \frac{p}{Q} = 168^\circ$$

Slot number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Elec. angle	0	168	-24	144	-48	120	-72	96	-96	72	-120	48	-144	24	-168



U_{FA} :



$$k_w = \frac{A \cdot (1 + 2 \cdot \sin(66) + 2 \sin(78))}{5A} = 0,956$$

for 3rd harmonic

$$k_w^3 = \frac{A \cdot (1 + 2 \cdot \sin(18) + 2 \cdot \sin(54))}{5 \cdot A} = 0,647$$

for 5th harmonic

$$k_w^5 = \frac{A \cdot (1 + 2 \cdot \cos(120) + 2 \cdot \cos(60))}{5A} = 0,2$$

Slot number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Top layer	A ⁺	A ⁻	A ⁺	C ⁺	C ⁻	C ⁺	C ⁻	C ⁺	B ⁺	B ⁻	B ⁺	B ⁻	B ⁺	A ⁺	A ⁻
Bottom layer	A ⁺	A ⁻	C ⁻	C ⁺	C ⁻	C ⁺	C ⁻	B ⁻	B ⁺	B ⁻	B ⁺	B ⁻	A ⁻	A ⁺	A ⁻

Q3

- Electric Loading : $\bar{A} = \frac{N_{\text{turn}} \cdot I \cdot Q}{\pi D_i}$, Q: # of slots

$I = J \cdot A_{\text{cu}}$, A_{cu} = copper area

$A_{\text{cu}} = k_{\text{wc}} \cdot A_{\text{slot}} = 0,6 \cdot \frac{\pi \cdot (R_{\text{so}}^2 - R_{\text{si}}^2)}{15} \cdot 0,4 \text{ mm}^2 \text{ per slot}$

R_{so} : outer radius of stator $R_{\text{so}} = 50 \text{ mm}$

$$R_{ro} = 0,62 \cdot 50 = 31 \text{ mm (given)}$$

$$R_{si} = R_{ro} + l_m + p = 31 + 4 + 1 = 36 \text{ mm}$$

* to keep the peak flux density close to 1,4 T, we approximately use 60% of the total area for copper winding.

$$A_{cu} = 60,6 \text{ mm}^2 \text{ per slot}, \quad J = 5 \text{ Arms/mm}^2$$

$$\Rightarrow I = 5 \cdot 60,6 = 303 \text{ Arms}$$

$$\bar{A} = \frac{15 \cdot 303 \text{ Arms}}{\pi \cdot 2 \cdot 0,036} = 20\,093 \text{ Arms/m}$$

- Carter's Coeff. $k_c = \frac{\tau_0}{\tau_0 - k b_1}, \quad k = \frac{b_1 / \delta}{5 + b_1 / \delta}$

$$\text{tooth pitch } \tau_0 = \frac{2\pi R_{si}}{N_s} = \frac{2\pi \cdot 36 \text{ mm}}{15} \approx 15 \text{ mm}$$

$$\text{slot opening } b_1 = 0,45 \cdot \tau_0 \approx 6,75 \text{ mm}$$

$$\delta = p + \frac{l_m}{\mu_r} = 1 + \frac{4}{1,05} = 4,81 \text{ mm} \quad \text{effective air-gap}$$

$$k = \frac{6,75 / 4,8}{5 + 6,75 / 4,8} = 0,22, \quad k_c = \frac{15}{15 - 0,22 \cdot 6,75} \approx 1,11$$

$$\text{effective air gap clearance: } \delta_e = k_c \cdot p = 1,11 \cdot 1 = 1,11 \text{ mm}$$

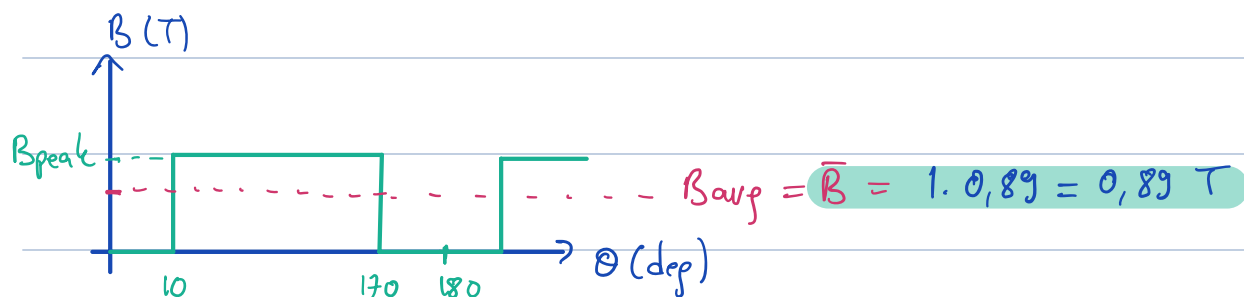
$$- L_{eff} = L_s + 2g = 100 + 2 = 102 \text{ mm}$$

- Air-gap peak flux density:

$$B_{peak} \approx \frac{B_r}{1 + \mu_r \cdot \frac{g}{l_m} \cdot k_c} = \frac{1.3}{1 + 1.05 \cdot \frac{1}{4} \cdot 1.11} \approx 1 \text{ T}$$

- Magnetic Loading:

$$\bar{B} = B_{peak} \cdot \alpha_{mp}, \quad \alpha_{mp}: \text{magnet-pole ratio (160/180)}$$



$$C = \frac{\pi^2}{12} \cdot k_w \cdot \bar{A} \cdot \bar{B} = \frac{\pi^2}{12} \cdot 0.956 \cdot 20093 \cdot 0.89 \approx 11931 \text{ J/m}^3$$

$\sigma = \bar{A} \cdot \bar{B} \rightarrow$ from Adelaide University paper, from lecture notes
or

$$\sigma = \frac{1}{12} \cdot k_w \cdot \bar{A} \cdot B_{peak} \rightarrow \text{ai suggestion.}$$

$$\sigma = \frac{1}{12} \cdot 0.956 \cdot 1 \cdot 20093 = 13582 \text{ N/m}^2$$

$$T = 2\pi \cdot R_{si}^2 \cdot L_{eff} \cdot \sigma = 2\pi \cdot 0.036^2 \cdot 0.102 \cdot 13582 = 11.28 \text{ Nm}$$

$$- \Phi = B_{av} \cdot \frac{2\pi R_{si} \cdot L_{eff}}{N_m} = 0.89 \cdot \frac{2\pi \cdot 0.036 \cdot 0.102}{14} = 1.466 \text{ mWb}$$

$$N_{ph} = \frac{V_{rms}}{4,44 \cdot f \cdot k_w \cdot \Phi} = \frac{V_{rms} \cdot \sqrt{2}}{\omega \cdot k_w \cdot \Phi}$$

Assume γ connected machine. $V_{rms} \cdot \sqrt{2} = \frac{48}{\sqrt{3}} = 27,7 V$

$$\omega: \omega_m = 1500 \cdot \frac{2\pi}{60} = 157 \text{ rad/s}$$

$$\omega_e = p \cdot \omega_m = 7 \cdot 157 = 1099 \text{ rad/s}$$

$$\Rightarrow N_{ph} \cong 18$$